Limit & Completions

...gical groups and inverse limit

V.1.1. G is a topological abelian group iff:

1) G is abelian;

2) G is a topological space;

3) $+: G\times G \to G$, $(x,y)\mapsto x+y$ and $-1: G \to G$, $x \mapsto x^{-1}$ are continuous.

Def V.1.2. G is first countable. The set of all Cauchy equivalence classes is denoted by $\hat{G}$, the addition on $\hat{G}$ is $+: (x_n)+(y_n)=(x_n+y_n)$.

Then $\hat{G}$ is called the completion of G

Remark. 1). $\hat{G}$ is an Abelian group. $\phi: G\to\hat{G}$ $a\mapsto(a)$ is a homomorphism.

and $\ker\phi = \bigcap U$. U is the neighbourhood of 0.

2). G, H are topological Abelian groups. $f: G\to H$ continuous & homomorphism

then $\hat{f}: \hat{G}\to\hat{H}$: $(x_n)\mapsto(f(x_n))$ is the homomorphism.

Lemma V.1.3. G is Hausdorff $\Leftrightarrow$ $\{0\}$ is closed in G.

Lemma V.1.4. $H := \bigcap_{\substack{0\in U \\ U \text{ open}}} U$, then i) H is a subgroup.

ii) $H = \overline{\{0\}}$.

iii) G/H is a Hausdorff Abelian group

iv). G Hausdorff $\Leftrightarrow H=\{0\}$.

pf: (skipped) ▨

Remark. 3). Suppose $G = G_0 \supseteq G_1 \supseteq \dots \supseteq G_n \supseteq \dots$ is a countable neighbourhood basis of 0. G_n are groups.

Then $G_n \subseteq G$ is closed and open subgroups.

Def V.1.5. Given a sequence of abelian groups $\dots \to A_{n+1} \xrightarrow{\theta_{n+1}} A_n \xrightarrow{\theta_n} \dots$

called this an inverse system

$\varprojlim A_n :=$ the group of all coherent sequences (a_n) $\left(\theta_{n+1}(a_{n+1})=a_n,\ a_n\in A_n\right)$

Remark. 3) pro: now the inverse system is $\dots \to \dots \to G/G_{n+1} \xrightarrow{\theta_{n+1}} G/G_n \xrightarrow{\theta_n} \dots$

where $\theta_{n+1} = \pi_{n+1}: G/G_{n+1} \to G/G_n \cong (G/G_{n+1})/(G_n/G_{n+1})$

and we have $\hat{G} \cong \varprojlim G/G_n$. (*)

Example $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$:

i) In algebra. $\mathbb{Z} \longleftarrow \mathbb{Z}/p\mathbb{Z} \leftarrow \mathbb{Z}/p^2\mathbb{Z} \leftarrow \dots \leftarrow \mathbb{Z}/p^n\mathbb{Z} \leftarrow \varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \{(a_n)_{n\geq 1} \in \prod_{n\geq 1} \mathbb{Z}/p^n\mathbb{Z} \mid a_n$

ii) In topology: take the topology on $\mathbb{Z}/p^n\mathbb{Z}$ as the discrete topology

Then $\mathbb{Z}_p$ is compact, disconnected, metric, with none isolated points

Thus $\mathbb{Z}_p \cong C$ where C is the Cantor set ▨

Prop. V.1.6. If $0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0$ is an exact sequence of inverse system,

then $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n$ exact.

Moreover, $\{A_n\}$ is surjective system., then $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$. exact.

pf: $A = \prod A_n$. B, C similarly. and $d^A: A \to A \quad (a_n) \mapsto (a_n - \theta_{n+1}(a_{n+1}))$

then $\operatorname{Ker} d^A = \varprojlim A_n$

Applying Snake-lemma on the following diagram:

$$\begin{array}{ccccccc} 0 & \to & A & \to & B & \to & C \to 0 \\ & & \downarrow d^A & & \downarrow d^B & & \downarrow d^C \\ 0 & \to & A & \to & B & \to & C \to 0 \end{array}$$

We have $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to \operatorname{Coker} d^A \to \operatorname{Coker} d^B \to \operatorname{Coker} d^C \to 0$.

If $A_{n+1} \xrightarrow{\theta_{n+1}} A_n$ surjective , ▨

given $a_n \in A_n$ $\exists\, x_{n+1} \in A_{n+1}$. s.t. $x_n - \theta_{n+1}(x_{n+1}) = a_n$ (by induction). $\Rightarrow d^A$ = surjective ▨

Cora V.1.7. $0 \to G' \to G \xrightarrow{P} G'' \to 0$ exact. G have the topology defined by $\{G_n\}$.

Then ang give G', G'' the induced topology. $\{G' \cap G_n\}$, $\{pG_n\}$.

Then $0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$ exact.

pf: notice that $0 \to \frac{G}{G' \cap G_n} \to \frac{G}{G_n} \to \frac{G''}{pG_n} \to 0$ exact. (It's easy to check).

Then by (*) in Remark 3) we have $0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$ exact.

Remark. let $G' = G_n$., $G'' = G/G_n$. then G'' has the discrete topology (Since the basis are $G/G_n \supseteq G_1/G_n \supseteq \dots \supseteq \{0\}$. are finite. and by Remark 3). We have $\bar{g} \in G_i/G_n$ are open $\Rightarrow$ G'' has the discrete topology).

Cora V.1.8. $\hat{G}_n \trianglelefteq \hat{G}$ and we have $\hat{G}/\hat{G}_n \cong G/G_n$ by taking inverse limits we have $\hat{\hat{G}} \cong \hat{G}$.

Take I as an ideal in A. $G = A$. $G_n = I^n$. then the topology defined on A is called the I-adic topology.

Remark. 1). I-adic topology is Hausdorff $\Leftrightarrow \bigcap I^n = (0)$.

2) $\phi: A \to \hat{A}$ is a continuous ring homomorphism, $\ker \phi = \bigcap I^n$.

Def V. 1.10. M is an A-module. $G = M$. $G_n = I^n M$. This defines the I-topology on M.

Lemma V. 1.11. (Hensel Lemma) Let (R, m) be a ~~m-adic~~ local ring that is complete with respect to m-adic topology, $f(x) \in R[x]$, Suppose $\exists\, x_0 \in R$ s.t. $\begin{cases} f(x_0) \equiv 0 \pmod m \\ f'(a_0) \notin m \ (f'(a_0) \text{ is a unit in } R) \end{cases}$.

Then $\exists!\ a \in R$. s.t. $f(a) = 0 \quad a \equiv a_0 \pmod m$

pf: 1. Existence: we construct a sequence $\{a_n\}_n \subseteq R$. s.t.

(i) $a_n \equiv a_0 \pmod m$.

(ii) $f(a_n) \in m^{n+1}$

(iii) $a_{n+1} \equiv a_n \pmod{m^{n+1}}$.

$n=0$ ✓. Assuming that we have constructed a_n

$\Rightarrow a_n \equiv a_0 \pmod m \Rightarrow f'(a_n) \equiv f'(a_0) \not\equiv 0 \pmod m$

Let $v = f'(a_n)^{-1}$ and let $a_{n+1} = a_n - f(a_n)v$. $\Rightarrow a_{n+1} \equiv a_0 \pmod m$ (i).

$\Rightarrow a_{n+1} - a_n = -f(a_n)v \in m^{n+1}$ (iii).

By ~~Factor~~ Taylor expansion over a ring:

we have $f(a_{n+1}) = f(a_n) + f'(a_n)(-f(a_n)v) + h^2 g(a_n, h) = h^2 g(a_n, h)$.

where $h = a_{n+1} - a_n \in m^{n+1} \Rightarrow h^2 \in m^{2n+2} \in m^{n+2}$. $\Rightarrow f(a_{n+1}) \in m^{n+2}$ (ii).

2. Convergence: $\forall\, m > n \geq 0$. we have $a_m - a_n = \sum_{i=n}^{m-1} (a_{i+1} - a_i) \in m^{n+1}$

$\Rightarrow \{a_n\}$ is a Cauchy sequence.

By the completeness. $\exists!\ a$ s.t. $a \equiv a_n \pmod{m^{n+1}}$. $\forall\, n$. $(a \equiv a_0 \pmod m)$

The poly-map is continuous $f(a) = f(\lim_{n\to\infty} a_n) = \lim_{n\to\infty} f(a_n) = 0$.

$\Rightarrow f(a) = 0$.

3. Uniqueness: $f(a) = f(b) = 0 \quad a \equiv b \equiv a_0 \pmod m$. $d = a - b \in m$.

$\Rightarrow 0 = f(a) = f(b) + f'(b)d + d^2 g(b, d)$ $(d = a - b \in m)$.

$\Rightarrow d = -d^2 g(b,d) f'(b)^{-1} \Rightarrow d(1 - dh) = 0$. $(d \in m)$

$\Rightarrow dh \in m$. $\Rightarrow 1 - dh$ is a unit in $R \Rightarrow d = 0 \Rightarrow a = b$ ▨